

Math 242, S13
Quiz 5

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the following initial value problem:

$$\frac{dP}{dt} = 2P(5 - P), \quad P(0) = 2.$$

(a) Solve the initial value problem for $P(t)$.

$$\begin{aligned}\frac{dP}{dt} &= 2P(5-P) \\ \frac{1}{P(5-P)} dP &= 2dt \\ \int \frac{1}{P(5-P)} dP &= \int 2dt \\ \frac{1}{5} \int \frac{1}{P} + \frac{1}{5-P} dP &= 2t + C \\ \ln|P| - \ln|5-P| &= 10t + C \\ \ln\left|\frac{P}{5-P}\right| &= 10t + C \\ \left|\frac{P}{5-P}\right| &= e^{10t+C} \\ \frac{P}{5-P} &= Be^{10t}\end{aligned}$$

$$\begin{aligned}P(0) = 2 &\Rightarrow \frac{2}{5-2} = Be^{10 \cdot 0} \\ \frac{2}{3} &= B\end{aligned}$$

$$\begin{aligned}\text{thus} \\ \frac{P}{5-P} &= \frac{2}{3}e^{10t} \\ P &= \frac{10e^{10t}}{3+2e^{10t}} \\ P(t) &= \frac{10}{2+3e^{-10t}}\end{aligned}$$

(b) Find the limit $M = \lim_{t \rightarrow +\infty} P(t)$.

$$\begin{aligned}\lim_{t \rightarrow +\infty} P(t) &= \lim_{t \rightarrow +\infty} \frac{10}{2+3e^{-10t}} \\ &= \frac{10}{2+3 \cdot 0} = \frac{10}{2} = 5\end{aligned}$$